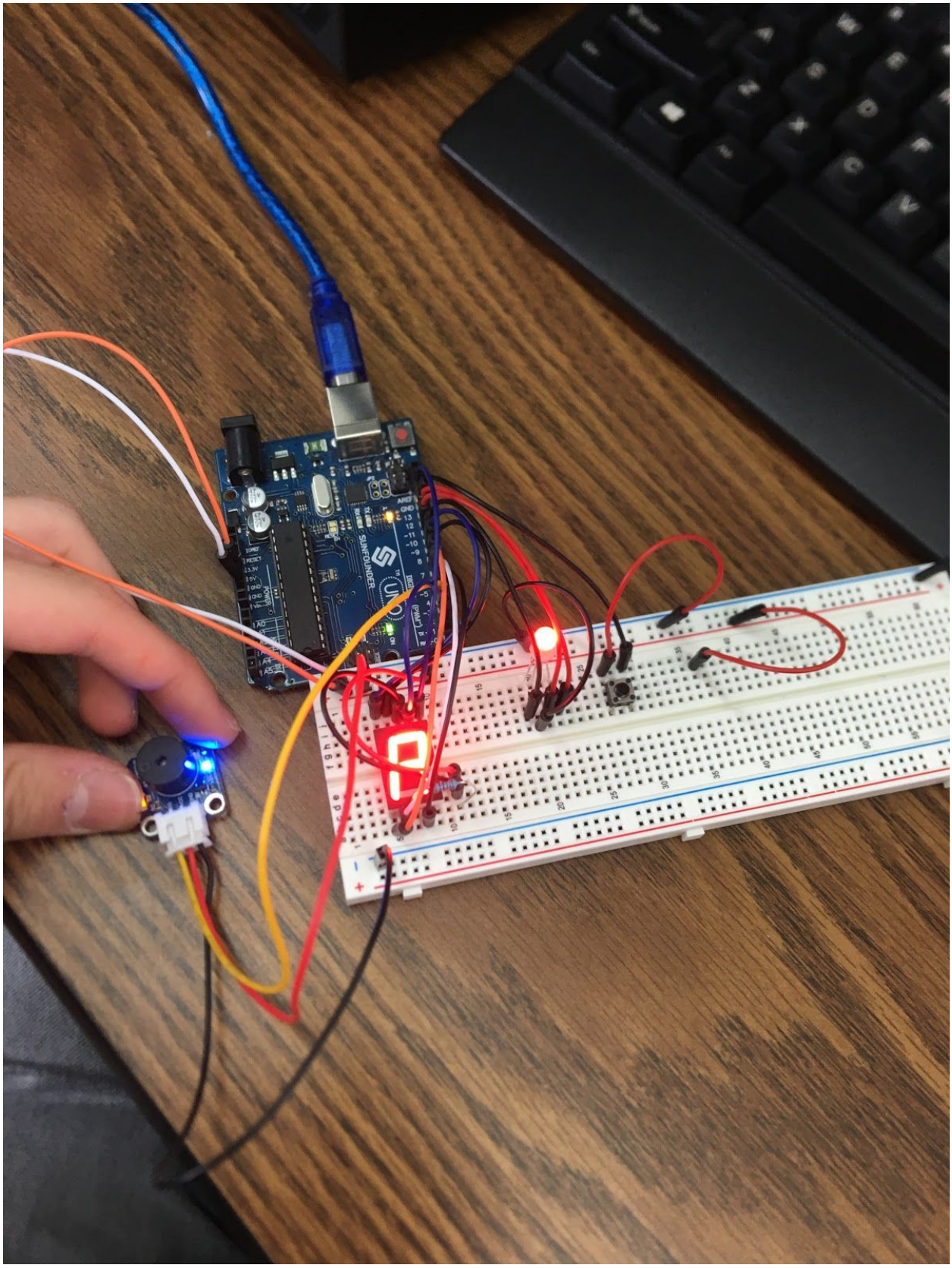
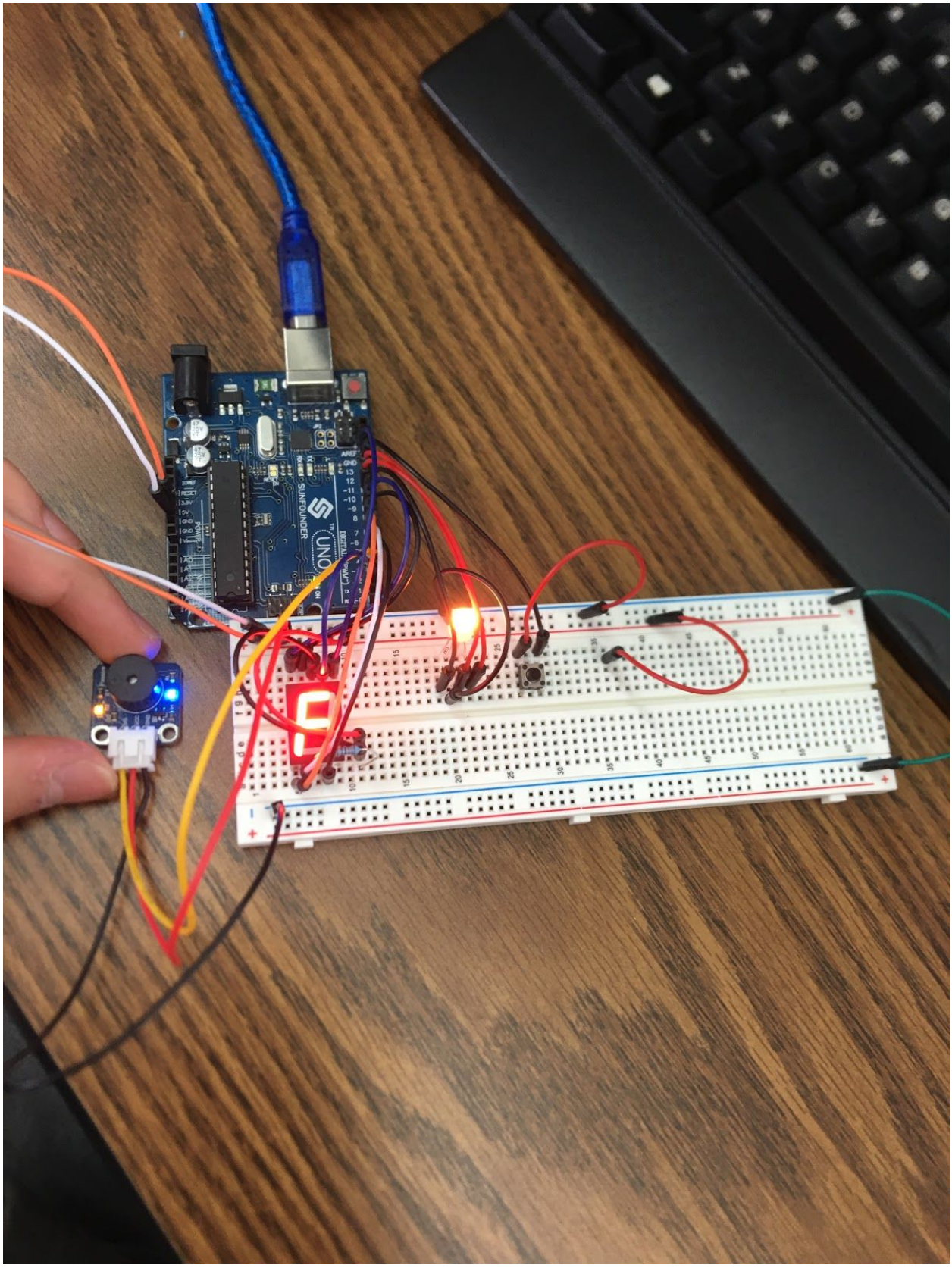
```
/*TITLE: ACTIVITY 2: CASE STRUCTURES - PUSH BUTTON
   DESCRIPTION: USE THE PUSH BUTTON TO DISPLAY LETTERS AND NUMBERS ON THE SEVEN SEGMENT DISPLAY
   DATE STARTED: 12/10/18
   FINISHED: 12/20/18
   DONIYOR NIMATULLO
*/

const int SLA = 2, SLB = 13, SLC = 4, SLD = 5, SLE = 6, SLF = 7, SLG = 8, rled = 9, gled = 10, bled = 11, pbport = 12, buzzer = 3
int pbvalue, pbcounter;

void setup()
{
  Initialize();
  analogWrite(rled, 0);
  analogWrite(gled, 0);
  analogWrite(bled, 0);
  pbvalue = digitalRead(pbport);
  Serial.println(pbvalue);
}

void loop()
{
  ButtonPress();
}
void Initialize()
{
  pinMode(SLA, OUTPUT);
  pinMode(SLB, OUTPUT);
  pinMode(SLC, OUTPUT);
  pinMode(SLD, OUTPUT);
  pinMode(SLE, OUTPUT);
  pinMode(SLF, OUTPUT);
  pinMode(SLG, OUTPUT);
  pinMode(rled, OUTPUT);
  pinMode(gled, OUTPUT);
  pinMode(bled, OUTPUT);
  pinMode(pbport, INPUT_PULLUP);
  pinMode(buzzer, 3);
}

void ButtonPress()
{
  pbvalue = digitalRead(pbport);
  if (pbvalue == 0)
  {
```

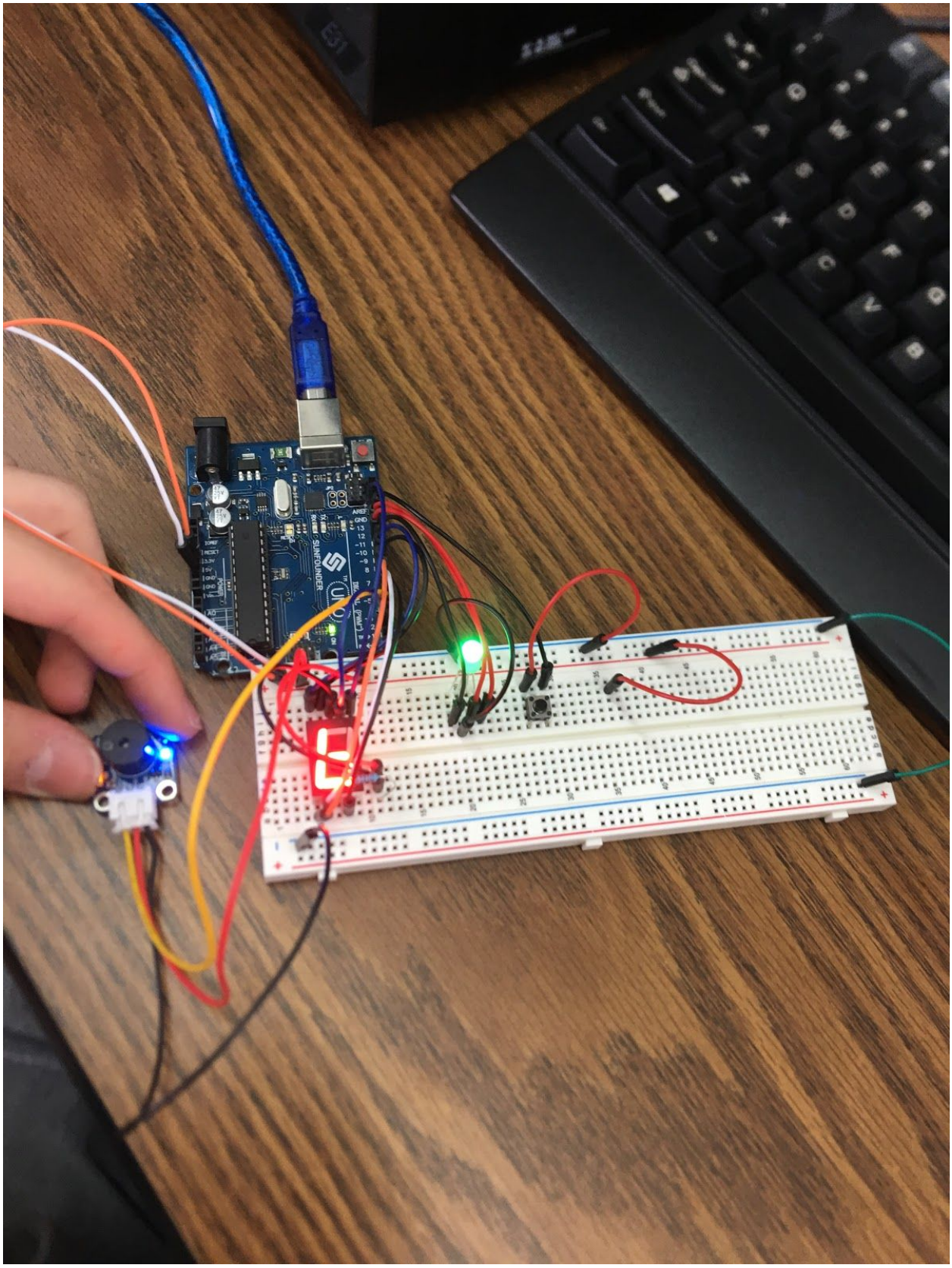
```
    LetterSequence(); //initiates sequence of letters on seven segment display
  }
}

void LetterSequence()
{
  switch (pbcounter)
  {
    case 1:
      {
        SevenSeg(1, 1, 1, 0, 1, 1, 1); //A
        analogWrite(rled, 30);  //first color value
        analogWrite(gled, 0);
        analogWrite(bled, 0);
        tone(buzzer , 100);
        delay(500);
        break;
      }

    case 2:
      {
        SevenSeg(0, 0, 1, 1, 1, 1, 1); //b
        analogWrite(rled, 0); //second color value
        analogWrite(gled, 30);
        analogWrite(bled, 0);
        tone(buzzer , 200);
        delay(500);
        break;
      }

    case 3:
      {
        SevenSeg(1, 0, 0, 1, 1, 1, 0); //C
        analogWrite(rled, 20); //third color value
        analogWrite(gled, 150);
        analogWrite(bled, 0);
        tone(buzzer , 300);
        delay(500);
        break;
      }

    case 4:
      {
        SevenSeg(0, 1, 1, 1, 1, 0, 1); //d
        analogWrite(rled, 250); //third color value
        analogWrite(gled, 30);
```

```
    case 3:
      {
        SevenSeg(1, 0, 0, 1, 1, 1, 0); //C
        analogWrite(rled, 20); //third color value
        analogWrite(gled, 150);
        analogWrite(bled, 0);
        tone(buzzer , 300);
        delay(500);
        break;
      }

    case 4:
      {
        SevenSeg(0, 1, 1, 1, 1, 0, 1); //d
        analogWrite(rled, 250); //third color value
        analogWrite(gled, 30);
        analogWrite(bled, 0);
        tone(buzzer , 400);
        delay(500);
        break;
      }
    case 5:
      {
        SevenSeg(1, 0, 0, 1, 1, 1, 1); //e
        analogWrite(rled, 30); //fourth color value
        analogWrite(gled, 30);
        analogWrite(bled, 30);
        tone(buzzer , 500);
        delay(500);
        break;
      }
    case 6:
      {
        SevenSeg(1, 0, 0, 0, 1, 1, 1); //f
        analogWrite(rled, 80); //fifth color value
        analogWrite(gled, 20);
        analogWrite(bled, 0);
        tone(buzzer , 600);
        delay(500);
        noTone(buzzer);
        break;
      }
    default:
```

```
        analogWrite(rled, 250); //third color value
        analogWrite(gled, 30);
        analogWrite(bled, 0);
        tone(buzzer , 400);
        delay(500);
        break;
      }
    case 5:
      {
        SevenSeg(1, 0, 0, 1, 1, 1, 1); //e
        analogWrite(rled, 30); //fourth color value
        analogWrite(gled, 30);
        analogWrite(bled, 30);
        tone(buzzer , 500);
        delay(500);
        break;
      }
    case 6:
      {
        SevenSeg(1, 0, 0, 0, 1, 1, 1); //f
        analogWrite(rled, 80); //fifth color value
        analogWrite(gled, 20);
        analogWrite(bled, 0);
        tone(buzzer , 600);
        delay(500);
        noTone(buzzer);
        break;
      }
    default:
      {
        break;
      }
  }
}

void SevenSeg(int a, int b, int c, int d, int e, int f, int g)
{
  digitalWrite(SLA, a);
  digitalWrite(SLB, b);
  digitalWrite(SLC, c);
  digitalWrite(SLD, d);
  digitalWrite(SLE, e);
  digitalWrite(SLF, f);
  digitalWrite(SLG, g);

```




SUNFOUNDER
UNO


SUNFOUNDER